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REVIEW ARTICLE

Type-3c Diabetes Mellitus, Diabetes of Exocrine Pancreas – An Update

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Abstract: Background: The incidence of diabetes is increasing steeply; the number of diabetics has doubled over the past three decades. *Surprisingly*, the knowledge of type 3c diabetes mellitus (T3cDM) is still unclear to the researchers, scientist and medical practitioners, leading towards erroneous diagnosis, which is sometimes misdiagnosed as type 1 diabetes mellitus (T1DM), or more frequently type 2 diabetes mellitus (T2DM). This review is aimed to outline recent information on the etiology, pathophysiology, diagnostic procedures, and therapeutic management of T3cDM patients.

Methods: The literature related to T3cDM was thoroughly searched from the public domains and reviewed extensively to construct this article. Further, existing literature related to the other forms of diabetes is reviewed for projecting the differences among the different forms of diabetes. Detailed and updated information related to epidemiological evidence, risk factors, symptoms, diagnosis, pathogenesis and management is structured in this review.

Results: T3cDM is often misdiagnosed as T2DM due to the insufficient knowledge differentiating between T2DM and T3cDM. The pathogenesis of T3cDM is explained which is often linked to the history of chronic pancreatitis, pancreatic cancer. Inflammation, and fibrosis in pancreatic tissue lead to damage both endocrine and exocrine functions, thus leading to insulin/glucagon insufficiency and pancreatic enzyme deficiency.

Conclusion: Future advancements should be accompanied by the establishment of a quick diagnostic tool through the understanding of potential biomarkers of the disease and newer treatments for better control of the diseased condition.

Keywords: Type 3c diabetes mellitus, insulin, glucagon, pancreatogenic diabetes, chronic pancreatitis, pancreatic ductal adenocarcinoma.

1. INTRODUCTION

Diabetes mellitus (DM) is an assorted cluster of diseases marked by chronic blood glucose level elevation. Affecting around 425 million people worldwide, diabetes is a global disease that is currently on the rise [1]. Diabetes was previously divided into three main types, namely, type 1 diabetes mellitus (T1DM), type 2 diabetes mellitus (T2DM) and gestational diabetes. However, the latest classification of diabetes mellitus by the American Diabetes Association and World Health Organization defines diabetes mellitus as a

result of pancreatic diseases as pancreatogenic diabetes or T3cDM [2]. The proposed universal terminology now is diabetes of the exocrine pancreas [3], although terms such as type 3c diabetes and secondary pancreatic diabetes are still used [4]. The pancreatic diseases referred to pancreatic exocrine damage due to acute, relapsing and chronic pancreatitis (of any aetiology), cystic fibrosis, hemochromatosis, pancreatic cancer, pancreatectomy, and rare causes such as neonatal diabetes due to pancreatic agenesis [2].

There is little comprehensive information on the prevalence of T3cDM. Some older studies estimated a low prevalence of around 0.5%-1.15% among all the diabetes mellitus cases in North America [5, 6]. Separate studies from Southeast Asia, where tropical or fibrocalcific pancreatitis is endemic, show a greater prevalence of nearly 15% - 20% of all

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cases of diabetes mellitus [2]. Recent reviews show a prevalence of 5% - 10% among all diabetic patients in Western populations, indicating that the condition is more common than generally thought. In almost four-fifths of all T3cDM cases, chronic pancreatitis seems to be the main underlying cause, suggesting that the real prevalence of (subclinical) chronic pancreatitis seems to be markedly underestimated [7].

Diabetes of the exocrine pancreas is thought to make up 9% of diabetes in hospitalized patients [8]. To the best of our knowledge, preceding pancreatic disease has not been systematically identified in primary care in people with diabetes, and thus, the incidence and clinical characteristics of diabetes of the exocrine pancreas remain uncertain. The condition is frequently misdiagnosed as T2DM with only 2.7% of cases being correctly diagnosed [4]. The median age at diagnosis for T3cDM was 59 years, and 58.9% were male; the median body mass index among these patients was 29.2 kg/m² [4]. Surprisingly, the prevalence of T3cDM was significantly higher than that of adult-onset T1DM, at 2.59 per 100,000 people/year versus 1.64 per 100,000 people/year, though still noticeably lesser than that of T2DM, at 142.89 per 100,000 people/year. These data show that this under-recognized form of diabetes is much more widespread than previously believed [4].

T3cDM is due to the direct influence of many diseases of the exocrine pancreas leading to hyperglycaemia [9]. Several studies have reported on the higher risk of developing T3cDM in patients diagnosed with acute pancreatitis. A study on 2966 individuals after acute pancreatitis was conducted and it was reported that new onset of diabetes is 2.5 times higher among the patients who had an acute pancreatitis attack [10]. In another similar study, Lee *et al.* have reported similar findings and the risk of new-onset diabetes is 2.1 times higher among the patients after acute pancreatitis. This study was conducted in 3187 individuals, those who had an acute pancreatitis attack [11]. Pendharkar *et al.* showed that the prevalence of diabetes in a total of 110,042 individuals with diseases of the exocrine pancreas (acute or chronic pancreatitis, or pancreatic cancer) is 0.11% [12].

In a study conducted on 1868 diabetes patients, 9.2% of the diabetic patients were identified as T3cDM. Among these T3cDM patients, 78.5% were diagnosed with chronic pancreatitis, 10% with pancreatic cancer, 9% with hereditary hemochromatosis and 5% with cystic fibrosis [8]. In addition to its endocrine effect, chronic pancreatitis can cause progressive destruction in pancreatic tissue which may lead to partial or complete loss of digestive enzyme and bicarbonate secretion [13]. Patients with inflamed and fibrous pancreas have a high risk of developing pancreatic ductal adenocarcinoma (PDAC) or pancreatic cancer which makes it the second most common cause of T3cDM that consists of 8% of the cases [8]. Significantly increased risk of PDAC in new-onset diabetes seems to be the cause of reverse causality [14]. This concept is verified by various clinical, epidemiological, and experimental evidence. In a study, 80% of the patients with PDAC showed unusual fasting glucose or glucose intolerance despite the stage of cancer [15-17]. After oral glucose tolerance tests, about 2/3rd of the patients suffering from PDAC were identified to have diabetes and the

prevalence is nearly 45% when fasting glucose was used to diagnose diabetes [15-18]. On the other hand, the first appearance of diabetes is normally associated to the diagnosis of PDAC [19]. There were about 75-88% of the patients stated that diabetes diagnosed less than 2-3 years before the diagnosis of PDAC [20-22]. Also, successful management of PDAC frequently improves hyperglycemia for those with new-onset diabetes due to pancreatic cancer, but improvement is not seen in those with long-term diabetes [14,16]. Pancreatic surgery is another risk factor for developing T3cDM. Patients who are able to restrain from pancreatic polypeptide (PP) deficiency after the surgery, they have a lower risk of developing postoperative diabetes.

Other risk factor contributing to the development of T3cDM is gender, where males are more likely to develop chronic pancreatitis and thus more prone to developing T3cDM. The long duration of pancreatic disease, smoking and the presence of pancreatic calcification increases the likelihood [23]. The common causes of T3cDM are listed as chronic pancreatitis, hereditary hemochromatosis, pancreatic cancer, post-pancreatic resection and cystic fibrosis. However, several factors are likely to have inflated the prevalence of the condition, including unresolved questions concerning whether certain abnormalities such as decreased faecal elastase-1 are a consequence of a disease of the exocrine pancreas or a secondary effect of diabetes [24]. However, the exact risk factor of T3cDM is yet to be determined [25]. The common risk factors contributing to the development of T3cDM and other types of DM are illustrated in Table 1.

2. PATHOPHYSIOLOGY

The pathophysiology of T3cDM is mostly due to the pancreatic inflammation and further irreversible fibrosis of islet cell that progresses to islet cell loss. In T3cDM, the damage not only involved β -cell mass but also damages the PP secreting cells in the early phase of the disease. In the late phase of the disease, it damages the α -cell of islets resulting decreased glucagon levels [26, 27]. The pathogenesis of T3cDM is different to the type-2 and type-1 DM where only the β -cell function is compromised and damaged. The pathophysiology of the disease is explained in this review by considering five major functional changes. They are insulin deficiency, insulin resistance, pancreatic immunopathogenesis, reduced incretin effect and genetic association with the disease.

2.1. Insulin Deficiency

Anatomically, blood flows through the capillaries from the endocrine pancreas to the exocrine pancreas that is surrounded the islets [28]. This "portal" system provides a high concentration of endocrine hormones (insulin, glucagon, amylin, somatostatin and PP) from islets to acinar cells of the exocrine pancreas. The significance of these hormones in exocrine pancreas is not completely understood but, the acinar cells express insulin receptors that are involved in the regulation of digestive enzyme synthesis in exocrine pancreas [29, 30]. Scarring of pancreatic tissue which in turn leads to insulin deficiency can be elucidated by the anatomical and functional interplay between pancreatic islets as well as acinar tissue [31].

Table 1. Differential characteristics of T1DM, T2DM and T3cDM.

-	T1DM	T2DM	T3cDM
Onset age of Diabetes	Childhood/adolescent	Adult	Childhood/adult
Risk factors	• Hereditary • Environmental factors (viral infection, lower Vit-D and lower UVB exposure) • Dietary and lifestyle factors (childhood obesity, rapid growth of infants, older maternal age and short duration of breastfeeding)	• Diet (high glycaemic index food and carbohydrate intake) • Sedentary lifestyle • History of gestational diabetes • Hereditary	Secondary to • Acute/chronic pancreatitis • Cystic fibrosis • Hemochromatosis • Pancreatic cancer • Pancreatectomy
Symptoms	• Hypoinsulinemia • Severe hyperglycaemia • Frequent hypoglycaemia • Ketacidosis	• Hyperinsulinemia/Insulin resistance in the early phase of the disease. • Hypoinsulinemia in the late phase of the disease. • Hyperglycaemia • Rare hypoglycaemia • Reduced peripheral and hepatic insulin sensitivity • Impaired glucose tolerance	• Hypoinsulinemia, PP and pancreatic enzyme deficiency in the early phase of the disease. • In addition to hypoinsulinemia, PP and pancreatic enzyme deficiency, decreased glucagon level is observed in the late phase of the disease • Hyperglycaemia • Frequent hypoglycaemia • Reduced hepatic insulin sensitivity • Impaired glucose tolerance • Maldigestion • Occasional steatorrhoea
Diagnosis	• Plasma glucose level ≥ 126 mg/dl (7 mmol/L) • HbA1c level $\geq 6.5\%$ • 2h postprandial glucose (75g glucose load) ≥ 200 mg/dl (11.1 mmol/L)	<i>Prediabetes</i> • Fasting plasma glucose level (100 to 125 mg/dl) • HbA1c level (5.7 to 6.4%) • 2h postprandial glucose (75g glucose load) ≥ 140- 199 mg/dl (7.8 to 11 mmol/L) <i>Diabetes</i> • Plasma glucose level ≥ 126 mg/dl (7 mmol/L) • HbA1c level $\geq 6.5\%$ • 2h postprandial glucose (75g glucose load) ≥ 200 mg/dl (11.1 mmol/L)	• Fasting plasma glucose level ≥ 126 mg/dl (7 mmol/L) • HbA1c $\geq 6.5\%$ • 2h postprandial glucose (75g glucose load) > 11 mmol/L In addition to the above glycaemic parameters, must have all the major criteria as below: <i>Major criteria</i> (Pancreatic insufficiency, Pathological pancreas, and absence of autoimmune markers of T1DM) <i>Minor criteria</i> (impairment of beta-cell function, absence of insulin resistance, impairment in incretin secretion and decreased concentrations of lipid soluble vitamins).
Pathogenesis	• The autoimmune disorder causing β-cell destruction at the early age of life causing insulin deficiency.	• Progressive insulin secretory defect on the background of insulin resistance. • Impairment in insulin signaling pathways and reduced expression of glucose transporter in insulin-sensitive tissues. • Increased blood glucose level due to insulin resistance is compensated by hyperinsulinemia in the early phase of T2DM. • In the late phase, insulin secretion reduced due to β- cell damage causing insulin deficiency.	• Inflammation, fibrosis in pancreatic tissue leading to damage. • Reduction in insulin receptor and glucose transporter (GLUT2) in hepatocytes causes hepatic insulin resistance. • In the early phase, β-cell and exocrine pancreatic cell damage cause hypoinsulinemia, PP and pancreatic enzyme insufficiency. • In the late phase, α-cell damage cause reduced glucagon level which contributed the hypoglycaemic episodes in T3cDM. • Reduced incretin (GIP) and PP levels due to islets damage. • Reduced pancreatic enzyme level due to exocrine tissue damage.

(Table 1) Contd...

-	T1DM	T2DM	T3cDM
Pharmacological management	Insulin	- Biguanides, - Sulphonylureas - Thiazolidinediones - DPP-4 inhibitors - GLP-1 analogues - Sodium-glucose cotransporter-2 (SGLT2) inhibitors α-glucosidase inhibitors - Insulin	Management of Hyperglycaemia - Biguanides (metformin) - Sulphonylureas - Thiazolidinediones - DPP-4 inhibitors - GLP-1 analogues α-glucosidase inhibitors - Insulin Management of pancreatic insufficiency - Pancreatic enzyme replacement therapy (PERT) - Vitamin-D supplementation

A cross-sectional study of chronic pancreatitis patients suggests a progressive insulin deficiency. A mild insulin deficiency was also observed in those patients even before the development of diabetes [32, 33]. In another cross-sectional study of patients with chronic pancreatitis but without diabetes, some of those patients demonstrated an impaired fasting glucose and glucose disposal index in the intravenous glucose tolerance test [34]. The principal mechanisms behind this observation can be due to the inflammation and fibrosis of exocrine pancreas that slowly destroys the pancreatic islets. First, in the early stage of chronic pancreatitis, high level of cytokines and stressful inflammatory environment leads to β -cell dysfunction [34]. Second, in chronic pancreatitis, the functional tissues of pancreas like pancreatic acini, ducts, and nerve bundles eventually lose their vascularity as they are replaced by connective tissue due to considerable fibrosis of the pancreatic parenchyma [31, 33]. As chronic pancreatitis progresses, the pancreatic islet tissues were gradually destroyed due to the extensive fibrosis of the exocrine pancreas [35]. Furthermore, since many patients with chronic pancreatitis experience exocrine PP, a theory has levied of nutrient maldigestion due to the destruction of PP-secreting cells that lead to impaired incretin secretion. Nevertheless, some studies have shown that impaired incretin effect is a result, not a cause, of the diabetic state [7, 36]. The pathogenesis of T3cDM is eventually because of reduced insulin secretion as a result of decreased number and efficacy of islets due to inflammation, fibrosis and sclerosis [26, 37].

The α -cell function in pancreatic islet is inversely regulated by β -cell in a paracrine mechanism. In the initial stage of the T3cDM, defects in insulin secretion due to β -cell damage is associated with increased glucagon secretion which might have involved impairment of glucose tolerance [38] and it was observed that pathologically there is an increased number of α -cells in islets [33]. With the progressive islets damage, glucagon production from pancreatic α -cells is also diminished in T3cDM, which may explain the episodes of severe hypoglycaemia that are reported in some patients [38, 39]. These pathological changes are different from the pancreas of T1DM where there is an autoimmune-mediated β -cell destruction [40]. Additionally, T2DM is characterized by the formation of amyloid which is not seen in chronic pancreatitis [31, 41].

Impaired insulin secretion and β -cell function are markedly reduced in PDAC, whereas insulin resistance was only moderately increased [42]. In a cell-line study of PDAC, microarray analysis detected adrenomedullin as a mediator for β -cell toxicity and inhibition of insulin secretion [43, 44]. Further, in a clinical study, it was reported that plasma adrenomedullin concentration is higher in patients with diabetes secondary to pancreatic cancer in comparison to healthy control [43]. Adrenomedullin was reported as the key mediator that is overexpressed in PDAC and promote the tumor aggressiveness [45-47]. Thus adrenomedullin promotes the invasive behaviour as well as cause hyperglycaemia in the microenvironment of the cancerous pancreas.

2.2. Insulin Resistance

In addition to insulin deficiency, insulin resistance is also observed in chronic pancreatitis patients. In the euglycaemic-hyperinsulinaemic clamp studies of patients diagnosed with both chronic pancreatitis and diabetes, the degree of insulin sensitivity was lower than the healthy subjects or type 1 diabetes patients [48, 49]. By contrast, another clamp studies demonstrated an increased insulin sensitivity type 3c diabetes patients with comparison to T1DM [50, 51]. Hepatic glucose production was assessed in T3cDM patients using a clamp study. It was observed that T3cDM patients have elevated basal glucose production compared with healthy controls [49]. Besides, chronic pancreatitis, acute pancreatitis also involved in the pancreatic injury. There are some observations that correlate between the extent of pancreatic damage and failure of insulin production [52, 53]. As compared to chronic pancreatitis, acute pancreatitis involves more insulin resistance and less failure in insulin secretion [54]. All these studies suggested decreased hepatic insulin sensitivity whereas, increased peripheral insulin sensitivity in T3cDM.

Hepatic insulin resistance has been demonstrated in patients with T3cDM secondary to pancreatic resection, chronic pancreatitis, PDAC, and cystic fibrosis [15, 35, 55]. The cause of persistent hepatic glucose production and isolated hepatic insulin resistance in T3cDM appears to be multifactorial, with a deficient PP response playing a key part. Insulin receptor expression is upregulated in the liver by PP,

Insulin receptor reduction can be observed in PP deficiency ultimately leading to hepatic insulin resistance in T3cDM [55]. The reduction in PP concentration is an important physiological difference between T3cDM and T1DM. Diminished insulin receptor availability in chronic pancreatitis can be reversed by PP administration [56-58]. Data from clamp studies in patients with PP deficiency and chronic pancreatitis showed that PP administration reversed the hepatic insulin resistance [55]. In a subsequent randomized, blinded, placebo-controlled study, a 72 h subcutaneous infusion of PP improved overall insulin sensitivity in PP deficient patients with T1DM and T3cDM [59]. PP deficiency is a common finding in all forms of T3cDM studied so far, and these studies show that PP is a glucoregulatory hormone that regulates hepatic insulin sensitivity.

In addition to altered insulin receptor availability, altered hepatic insulin function in chronic pancreatitis has also been linked to the inflammation-based activation of hepatocyte I- κ B kinase- β and NF- κ B [60]. Interestingly, blockade of NF- κ B activation results in improved hepatic insulin sensitivity [61]. This effect can be achieved by activation of peroxisome proliferator-activated receptor- γ , which is the mechanism of action of the thiazolidinedione (TZD) class of antidiabetic drugs. Rosiglitazone has been shown to reverse hepatic insulin resistance in rats with chronic pancreatitis but has not been rigorously studied in human beings [14].

2.3. Immunopathogenesis

Immunopathogenesis contribute to the development of T3cDM. A study has demonstrated that pro-inflammatory cytokines like interleukin-1 β , tumour necrosis factor (TNF) α , and interferon- γ can inhibit glucose-stimulated insulin [8,61]. High concentration of interleukin-1 receptor and interleukin-1 β can induce β -cell apoptosis while a high concentration of interferon γ can lead to diminished translocation of a transcription factor known as pancreatic and duodenal homeobox 1 (PDX1) or insulin promoter factor 1 [14, 62]. PDX1 is essential for the development of pancreatic cells for examples β -cell maturation, and duodenal differentiation. In the patient with chronic pancreatitis, the loss of PDX1 reduces β -cell differentiation as well as the survival rate of β -cell (Fig. 1) [63]. As a result, the production of insulin is impaired. Furthermore, adrenomedullin and vanin-1, which are also being produced as a consequence of inflammation might alter the function of islet cells in T3cDM [64]. However, more studies are required to further investigate the exact role of these inflammatory mediators in T3cDM.

T3cDM secondary to PDAC seems to be different from any others causes of T3cDM as it is not because of fibro-inflammatory destruction or pancreatectomy but the paraneoplastic effect that caused by the mediators released by cancer [14, 65]. Nevertheless, the exact mechanisms of diabetes remain unclear, with evidence proposing that cancer

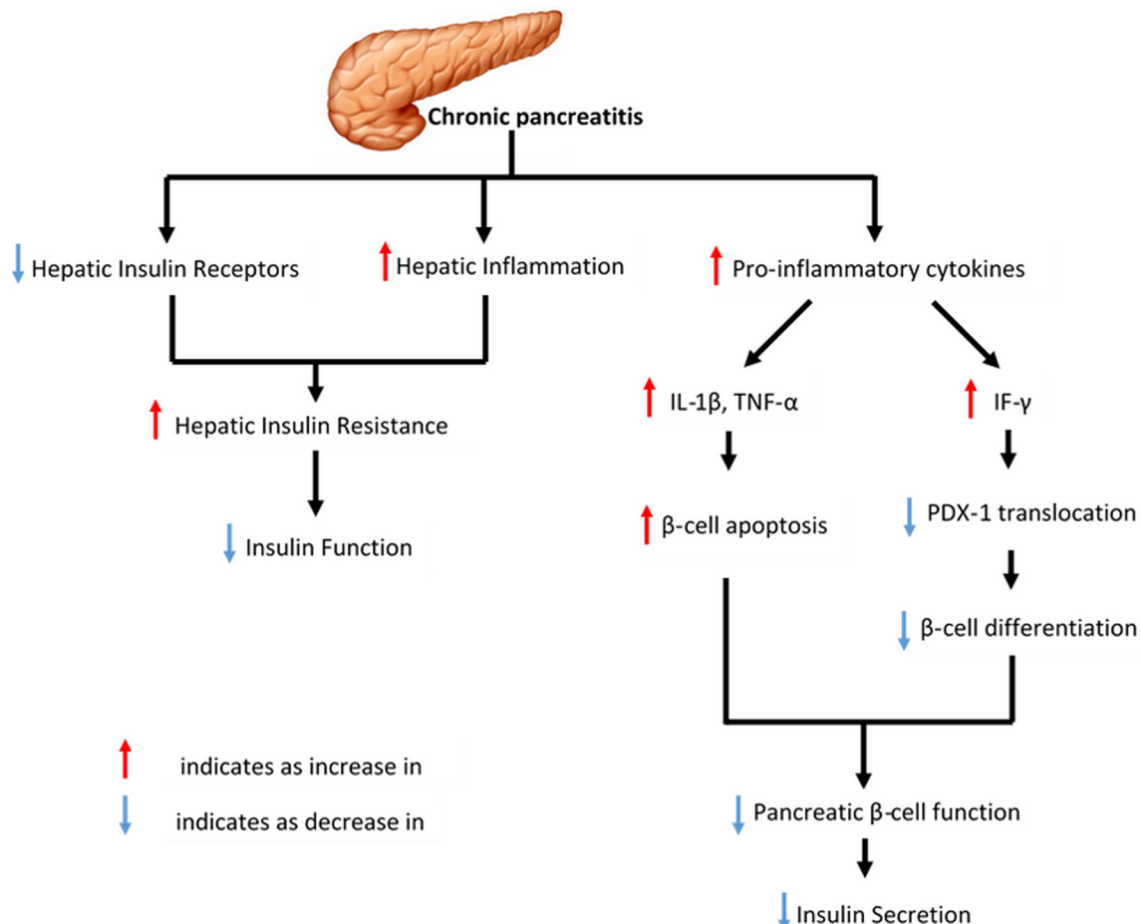


Fig. (1). Mechanism of altered insulin function and release that can lead to T3cDM in chronic pancreatitis patients. IL-1 β : Interleukin-1 β ; IF- γ : Interferon- γ ; TNF- α : Tumour necrosis factor- α ; PDX-1: Pancreatic and duodenal homeobox 1

resection in patients with PDAC improves both insulin resistance and β -cell dysfunction [14]. In several studies carried out by Permert and colleagues, there is reduced glucose metabolic capacity in patients with PDAC. This effect is more noticeable in patients with diabetes, but it also can be seen in those with a normal blood glucose level with PDAC [17, 18, 42, 66, 67]. Besides, investigators found that the level of plasma islet amyloid polypeptide was increased in patients with PDAC and diabetes, and they proposed that it may be part of the pathogenesis. However, the related studies failed to support this concept, therefore no further study has been conducted [21, 68].

Studies on PDAC patients revealed the presence of inflammation or high levels of calproteins (S100A8/A9) and connexin in pancreatic tissues [27, 69]. Overexpression of these proteins in islets of PDAC patients was observed [70]. A large quantity of S100A8/A9 can be found in neutrophils and activated macrophages, and this may explain the accumulation of monocytes, macrophages, neutrophils and other cells at early stages of pancreatic cancer [71, 72]. Based on a study using myoblast which is exposed to this peptide, attenuated the glucose utilization. Thus, the N-terminus of the S100A8 peptide has been suggested as a possible causative factor for diabetes [73]. Another research also found the upregulation of S100A9 in pancreatic tissue of PDAC with diabetes in comparison with healthy participants [74]. A study involved the peripheral blood mononuclear cell gene profiling for PDAC suggested that vanin-1 and matrix metalloproteinase-9 (MMP9) can be the potential biomarkers to differentiate between diabetes due to pancreatic cancer and T2DM. A later study showed that mRNA also associated with increased expression of MMP9, but not S100A8 [75]. With diabetes due to pancreatic cancer, MMP9 and S100A8 expression in macrophages are increased in presence of TNF α and interleukin-1 β that are abundant in a tumour microenvironment [76-78]. This inflammatory pathway somehow explains the impaired function of β -cell seen in patients with PDAC [61].

2.4. Reduced Incretin Effect

Glucagon-like peptide (GLP)-1 and glucose-dependent insulinotropic polypeptide (GIP) are the gut hormones that released in digestion and absorption of nutrients. GLP-1 secreted from L-cell of the intestine has many functions in gastrointestinal system that regulates insulin secretion and retards the gastric emptying. Importantly, GLP-1 is a principal gut hormone which stimulates insulin release, suppresses glucagon secretion, reduces appetite, and decelerates gastric emptying in response to increased body glucose [79, 80]. Furthermore, it stimulates the β -cell growth, neogenesis and differentiation which was reported in rodent and tissue culture study. In an *in vitro* study, GLP-1 showed inhibition of β -cell apoptosis [81]. Thus, GLP-1 stimulates β -cell proliferation and survival [82]. GIP secreted from α -cell of small intestine, augments glucose-stimulated insulin release from β -cell whereas, inhibits glucagon secretion from α -cell. Both these hormones are rapidly inactivated by dipeptidyl peptidase 4 (DPP-4).

In patients with steatorrhea due to alcoholic pancreatitis, GIP secretion is reduced which is normalized after enzyme

replacement therapy [83]. In another study, impaired GLP-1 secretion was reversed by pancreatic enzyme supplementation in patients with pancreatic insufficiency due to cystic fibrosis [84]. Pancreatic insufficiency may lead to the malabsorption of nutrient and decrease the release of incretin (GLP-1 and GIP) in response to body glucose thus, reduced the postprandial insulin secretion. However, some researchers concluded that reduced incretin release a consequence not a cause of T3cDM [83, 85, 86].

2.5. Genetic Association with T3cDM

Genetic association with T3cDM is concurrently related to the genetic predisposition in chronic pancreatitis and pancreatic cancer/PDAC. After several decades of research, a number of genes associated with chronic pancreatitis, diabetes mellitus and PDAC are identified [25]. Research suggests that pathogenesis of DM is explained by polygenic risk model where each genetic mutation has a small effect. Further, the risk of developing DM is increased in patients with common risk factors such as family history, age, sex, body mass index (BMI) and clinical conditions [87]. Along with these risk factors, smoking and alcohol intake may interact with genetic factors.

Studies have identified the specific genes for chronic pancreatitis. These are trypsinogen genes such as PRSS1, SPINK1, CFTR, CTSC, and CASR. Mutation in PRSS1 gene augments trypsinogen autoactivation, which explains the young age chronic pancreatitis [88]. Mutation in exon regions of PRSS1 has been studied so far to link with chronic pancreatitis and DM. In another study, abnormalities in the intron region of PRSS1 was linked to chronic pancreatitis but not to diabetes in the Chinese population [88]. The median age of onset of diabetes in PRSS1 gene exon and intron mutation was reported to be 36.6 years and 54.6 years respectively. Exon abnormality has a 60% risk of developing diabetes in a chronic pancreatitis patient [88]. The majority of patients with recurrent pancreatitis have multiple genetic variances and/or their interactions with environmental stressors [27].

PDAC resembles like chronic pancreatitis and possesses genetically heterogeneous. Mutation in CFTR, BRCA1, BRCA2, CDKN2A, PALB2 and ATM genes was reported earlier. [89-91]. Pancreatic Cancer Genetic Epidemiology Consortium (PACGENE), Pancreatic Cancer Case-Control Consortium (PanC4), and PanScan have demonstrated individuals have a higher risk of developing pancreatic cancer who has a family history of pancreatic cancer [92,93]. A number of vulnerable genes have been identified [94]. BRCA2 and PALB2 is the most commonly mutated gene in hereditary pancreatic cancer [89]. Mutation in the CFTR gene has a moderate risk in developing pancreatic cancer in young age [95]. Further, ATM mutation was linked to the familial pancreatic cancer predisposition [90].

3. DIAGNOSIS

Diagnosis of DM in patients with chronic pancreatitis may not be difficult, yet to classify T3cDM is often misleading. This is always due to underdiagnosed or misdiagnosed and often misclassified as T2DM [8]. To diagnose T3cDM, the initial evaluation is glycated haemoglobin (HbA1c) and

fasting glucose levels [96]. According to American Diabetes Association (ADA) the recent criteria of a fasting plasma glucose level of ≥ 126 mg/dl (7 mmol/l) or HbA1c of ≥ 48 mmol/mol (6.5%) indicates diabetes [96]. However, further evaluation is required when either one of these is impaired. Studies have also suggested distinguishing between T2DM and T3cDM through analysis of C-peptide and insulin levels [93]. The differential diagnostic parameters are listed in Table 1.

According to Ewald and Bretzel, to confirm if a patient is diagnosed with T3cDM, some major criteria must be present and they are a) presence of exocrine pancreatic insufficiency by monoclonal faecal elastase-1 test; b) pathological pancreas by endoscopic ultrasound, CT scan, or MRI and c) absence of related autoimmune markers of T1DM. There are also some minor criteria that may or may not present in patients. For example, impairment of β -cell function by assessing homeostatic model for glucose concentration or C-peptide or β -cell, absence of insulin resistance by assessing homeostatic model for insulin resistance, impairment of incretin secretion by testing GLP-1 or PP levels, and decreased in serum concentrations of lipid soluble vitamins (A, D, E, and K). As a result of low lipid soluble vitamin, symptoms of steatorrhea is present, but this can be easily reversed by pancreatic enzyme supplements [97]. The occurrence of ketoacidosis in T3cDM is rare, hyperglycemia is typically relatively mild, yet hypoglycaemia is common. T3cDM showed an increase in peripheral insulin sensitivity due to the absence of counter-regulatory mechanism of glucagon whereas decreased in hepatic insulin sensitivity due to relative hypoinsulinemia and deficiency in the PP secretion. Also, impairment of incretin secretion due to the malabsorption of protein and fat was observed. The [39,98].

4. COMPLICATIONS

The prevalence of diabetes-related complications in T3cDM is believed to be rather alike to that of T1DM and T2DM, which includes retinopathy, nephropathy, neuropathy along with microangiopathic complications. Due to the poor awareness of this condition, patients may be misdiagnosed, typically as T2DM [99]. The consequences of this misclassification delays in receiving appropriate treatments, which could lead in progressing the eye, nerve and kidney impairment [4]. It is generally accepted that macrovascular complications are not as prevalent in T3cDM due to poor dietary intake and pancreatic exocrine insufficiency; nevertheless, the data are uncertain and further continuous research is necessary to properly analyse the risks [99].

Information on acute metabolic complications, for example, diabetic ketoacidosis and hypoglycaemia, in pancreatic disorders are limited. In a study of thirty-six diabetic patients with chronic pancreatitis, insulin treatment resulted hypoglycaemia and severe hypoglycaemia in 78% and 17% of the patients respectively [100]. Information on traditional target organ complications due to T3cDM is also limited. There is only one prospective study so far, where fifty-four patients with T3cDM because of chronic pancreatitis or total pancreatectomy were studied for target organ complications. About 31% of the patients developed diabetic retinopathy which is correlated with the duration of diabetes [101]. In

other studies, 37%, 29% and 26% of the diabetic patients secondary to pancreatic disease reported to have retinopathy, nephropathy, and peripheral arterial disease respectively [102, 103]. However, these studies were published several decades ago with variable study designs. Thus, extrapolations of these available data are not reliable. Other factors such as hyperlipidaemia, hypertension, smoking and obesity also contribute in target organ complication which should be studied alongside with T3cDM [9].

Owing to the fact that both diabetes mellitus and chronic pancreatitis are risk factors for PDAC, this combination may increase concern for progression to PDAC [104]. Based on a population-based cohort study done in Taiwan, the risk of PDAC was considerably raised in patients with a combination of diabetes and chronic pancreatitis [105]. Other database studies exhibited a bigger, but more modest, risk ratio of PDAC in patients with otherwise unclassified diabetes and a history of chronic pancreatitis [106, 107]. However, additional validation from other studies is needed to confirm the complications of T3cDM.

5. MANAGEMENT

Due to misclassification, T3c diabetes patients often mismanaged as T2DM, which could result in poor outcomes of patients suffering from such condition. Currently, there is no specific guideline to address hyperglycemia or other related conditions in T3cDM. In addition, lack of comparative studies on the safety and efficacy of each individual antidiabetic agent in treating T3cDM also limits the development of guidelines to manage the disease at the grass root level. Nevertheless, a detailed understanding of the disease progression is somewhat helpful for the clinicians to decide what antidiabetic agents to control the symptoms in the affected patients, subsequently prevent deterioration of the disease. Generally, hyperglycemia and exocrine insufficiency should be tightly controlled, while nutritional management **also plays a significant role** in the management of T3cDM.

5.1. Managing Hyperglycemia

5.1.1. Metformin as a First-line Antidiabetic Drug

Metformin, an insulin sensitizer, is recommended to be used as first-line therapy in most patients with T2DM [108]. It is known to lower hepatic glucose output and raise peripheral insulin sensitivity. For the early management of diabetes secondary to chronic pancreatitis, metformin can be the drug of choice when hyperglycemia is mild (HbA1c $< 8\%$) [9, 109]. There are several supporting evidence towards the use of metformin as the first-line therapy in T3c diabetes patients. First, metformin can decrease the amount of insulin needed daily to lower the blood glucose level [110]. Second, an added benefit can be elaborated through the antineoplastic effects of metformin or its potential ability to reduce cancer risk in patients with diabetes secondary to pancreatic cancer [111-113]. Li *et al.* reported a reduction in the risk of pancreatic cancer by as much as 70% with metformin use although the meta-analysis conducted by Singh *et al.* does not support such association [112, 114]. Third, results from an animal study done by Maida *et al.* showed an increase in GLP-1 level in mice when metformin is administered, and

additionally enhanced pancreatic gene expression of GLP-1 and GIP receptors in the mice [115]. These effects can have a favourable impact on insulin secretion from pancreatic islets. Fourth, metformin is reported to be capable of prolonging the survival of diabetic patients with non-metastatic pancreatic cancer (32% lower mortality risk) despite the need to carry out further independent studies [116].

Although metformin seems advantageous in normalizing blood glucose levels, cautions must be taken against its side effects, one of which is lactic acidosis, which is potentially life-threatening in patients with pancreatogenic diabetes [117]. Gastrointestinal disturbances such as nausea, vomiting, and diarrhoea could be troublesome for patients with T3cDM on long-term use of metformin. Therefore, the dose of metformin should carefully be titrated to optimise therapy once it is initiated [118]. Another important consideration of metformin use is that it is contraindicated in patients **who** are alcoholic or susceptible to metabolic acidosis [117]. However, as long as the use of metformin is not contraindicated and is tolerable in terms of side effects, it can be regarded as primary therapy of hyperglycemia in patients with T3cDM [119].

5.1.2. Other Antidiabetic Agents

Initiation of oral antidiabetic agents with periodic dose adjustment is recommended if fasting blood glucose (more than 7.0 mmol/L) or 2h postprandial glucose is > 11.1 mmol/L after a challenge of 75 g oral glucose or HbA1c level > 7% [39]. The commonly prescribed drugs such as metformin, TZDs, α -glucosidase inhibitors (AGIs), sulfonylureas (SUs) and dipeptidyl peptidase-4 (DPP-4) inhibitors in the treatment of T2DM are included in the therapy of T3cDM. If metformin alone is not sufficient to control hyperglycemia, a TZD or an AGI can be added to achieve more effective glycemic stabilization [39]. As mentioned earlier, the side effects of these agents may prohibit their use in the T3cDM treatment. TZDs are known to carry risks of osteoporosis, being the most concerning side effect besides increasing risks of edema and congestive cardiac failure, are best avoided in patients with diabetes secondary to chronic pancreatitis who suffer from the risk of bone fracture [9,36,120]. Little or no evidence is discussed about the use of AGIs as added therapy. Furthermore, SUs, preferably short-acting ones for irregular meal intake, can be used in the treatment of early T3cDM as they promote insulin secretion although they carry the risk of hypoglycaemia [36, 109]. However, SUs seem to intensify the risk of mortality of cancer in diabetic patients in some studies, although others fail to prove any effect on risk of PDAC with SUs [25, 112]. Use of DPP-4 inhibitors is also controversial, very much similar as SUs, where the results about the elevated risk of pancreatic cancer are not consistent [25]. SUs and DPP-4 inhibitors can preferably be used in the treatment of diabetes with chronic pancreatitis than pancreatic cancer [98]. Moreover, evidence on the use of oral hypoglycaemic agents (OHAs) other than metformin is too little and unclear. Because of inadequate clinical studies on the safety and efficacy of individual antidiabetic drugs, use of these agents can only be continued upon monitoring their adverse effects, which are the major concern to the optimization of the antidiabetic regimen for patients with T3cDM.

5.1.3. Insulin Therapy

Insulin therapy is often given to patients with diabetes secondary to chronic pancreatitis to overcome insulin insufficiency [109]. It is preferably used in patients who have fasting blood glucose level (> 10 mmol/L) and HbA1c levels (> 8.5%), with weight loss and glycosuria [39]. The goal of insulin therapy is to normalize fasting blood glucose and HbA1c levels to 3.9 – 7.2 mmol/L and <7% respectively [119].

When OHA alone fails to correct hyperglycemia, insulin can be added for further control of such condition. However, patients with T3cDM might experience the risk of hypoglycaemia due to extra peripheral sensitivity to insulin [39]. Therefore, clinicians have to be judicious about the dose of insulin administered in such patients. Usually, the patients require less insulin compared to other diabetic patients, (normal dose is 10 U or 0.2 U/kg) at bedtime with an increment on three-day basis following readings on glucose estimation [108], who are dependent on insulin. Besides that, adjuvants may be added to insulin therapy to benefit patients with T3cDM with regards to reduced insulin requirement. Metformin has been proved to lower the daily requirement of insulin apart from other possible adjuncts such as glucosidase inhibitors [110-122].

Although chronic pancreatitis results in gradual deterioration of pancreatic islets functions which eventually mandates the use of insulin, there is a risk of pancreatic cancer/ductal carcinoma associated with its use [110, 112, 123]. There is also some concern regarding an increased risk of cancer mortality among insulin users with T2DM even though the studies are not conclusive [123].

5.2. Managing Exocrine Insufficiency

5.2.1. Pancreatic Enzyme Insufficiency

Exocrine dysfunction of the pancreas could occur in patients with T3cDM, leading to malabsorption and other metabolic complications [97]. However, clinical manifestations can usually be observed when there is more than 90% of exocrine pancreatic functions have declined [7].

To overcome the difficulties in protein and lipid digestion, pancreatic enzyme replacement therapy (PERT) should be initiated in which pancreatic enzymes should be administered together with each meal and monitoring of their therapeutic efficacy should be continuous [39]. For instance, pancreatin helps in improving response to insulin in patients with pancreatogenic steatorrhoea as well as reverses the decreased response of GIP [124]. Besides that, malabsorption of fat-soluble vitamins could also be correlated with pancreatic exocrine dysfunction. Vitamin D supplementation is important in terms of prevention of metabolic bone disorders because most patients with chronic pancreatitis would experience a deficiency in vitamin D [125].

5.3. Nutritional Management

Initially, PancreasFest Working Group has given a diagnostic and management approach to T3cDM in chronic pancreatitis [109]. They have suggested that T3cDM patients should be treated with medical nutrition that is tailored spe-

cifically to treat or prevent malnutrition, meal-induced hyperglycemia and control steatorrhoea in T3cDM patients. Andersen and Cui have mentioned that management with lifestyle modification should be initiated immediately after the diagnosis of T3cDM. In addition to nutritional management, weight reduction in obese patients, regular exercise, limiting the carbohydrate intake, alcohol and smoking cessation can benefit the patients those are having high blood glucose and cancer [39]. The dietary priority should be revised to minimize the frequency and degree of high blood glucose level in order to curtail the risks associated with the consequences of diabetes [126]. The consumption of drinks with sugar should be avoided especially unless treating hypoglycemic events. Preferentially, a low-glycemic index diet should be consumed. 'Diabetic food' or foods marketed especially for diabetic patients are not recommended generally as they may have the laxative effect if consumed in excess and are expensive [99].

A review done by Duggan *et al.* has focused on the detailed dietary management guideline for T3c diabetes patients. At first, management of diet should give priority to the prophylaxis of hypoglycemic events which occurs frequently in patients. Educating patients regarding the signs, symptoms and treatment of hypoglycemia could prevent the episodes of hypoglycaemia (Table 2) [99]. Regular intensive monitoring of blood glucose is suggested especially for those on intensive insulin regimens as recommended by ADA guidelines for T1DM and T2DM, but not patients with T3cDM [127]. However, these recommendations can be extrapolated for T3cDM patients who are on the intensive insulin regimens [99]. Those especially with 'brittle diabetes' should have continuous glucose monitoring and recorded a detailed level of blood glucose, dietary intake, pancreatic enzyme replacement therapy (PERT) usage and exercise (Fig. 2) [99].

Table 2. Principles of management for T3cDM in chronic pancreatitis [99].

Principles of Management	Management Plan
Prevent / Avoid - Low blood glucose level - High blood glucose level - Malabsorption - Worsening of malnutrition - Co-morbidities associated with diabetes	✓ Avoid skipping meals ✓ Have small, frequent meals ✓ Regular meal pattern together with regular starchy carbohydrates ✓ Glucose levels are measured frequently (if on insulin therapy, after physical activity, if poor diet, and if occur any symptoms of hypoglycaemic events) ✓ Reduce high-glycaemic index food and high sugar intake ✓ Ensure adequate Pancreatic enzyme replacement therapy (PERT) ✓ Regular dietician assessment and monitoring ✓ Record diet plan, PERT, exercise, glucose levels in a diary ✓ Stop alcohol intake and smoking



Fig. (2). Suggested self-monitoring regimen for testing blood glucose in T3cDM patients [128].

CONCLUSION

T3cDM is a type of diabetes mellitus that is often related to the pancreatic diseases which refer to the acute or chronic pancreatitis, pancreatic cancer, cystic fibrosis, pancreatic surgery, and hemochromatosis. T3cDM is frequently misdiagnosed or underdiagnosed as T2DM showing the similar laboratory results as T2DM. Hence, further evaluation of several parameters needs to be done in order to differentiate T2DM and T3cDM. Monitoring the abnormalities of the pancreatic function need to be done if the patients are suspected to have pancreatogenic diabetes. The complication of T3cDM is similar to that of T1DM and T2DM. Microvascular conditions are often presented in T3cDM patients rather than the macrovascular conditions. This is because of poor dietary intake and pancreatic exocrine insufficiency in T3c diabetes patients. Pharmacological management of T3cDM is mainly focused to treat the hyperglycemia and exocrine insufficiency in patients. Further, nutritional management has similar importance like pharmacological treatment in T3cDM patients.

Currently, there are limited findings and studies on T3cDM as compared to other types of diabetes mellitus. Hence, there is a limitation while diagnosing and treating T3cDM due to its poor understanding by the clinicians as well as little awareness among the public. Therefore, more studies and details findings should be carried out in order to increase the knowledge on T3cDM and thus establish the guidelines to diagnose and treat the patients with T3cDM.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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